

Calculus II

Lecture 10

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Applications of Definite Integrals.

- Definite Integral.
- Areas Between Curves.
- Multiple Integrals.

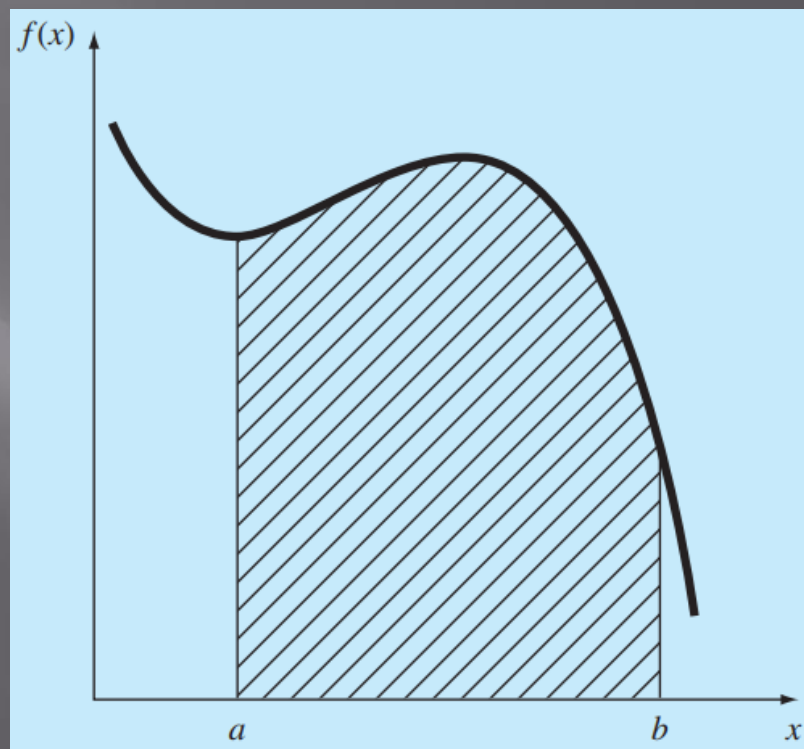
Definite Integral (1/13)

What is Definite Integral?

The process of measuring the area under a function plotted on a graph.

$$I = \int_a^b f(x) dx$$

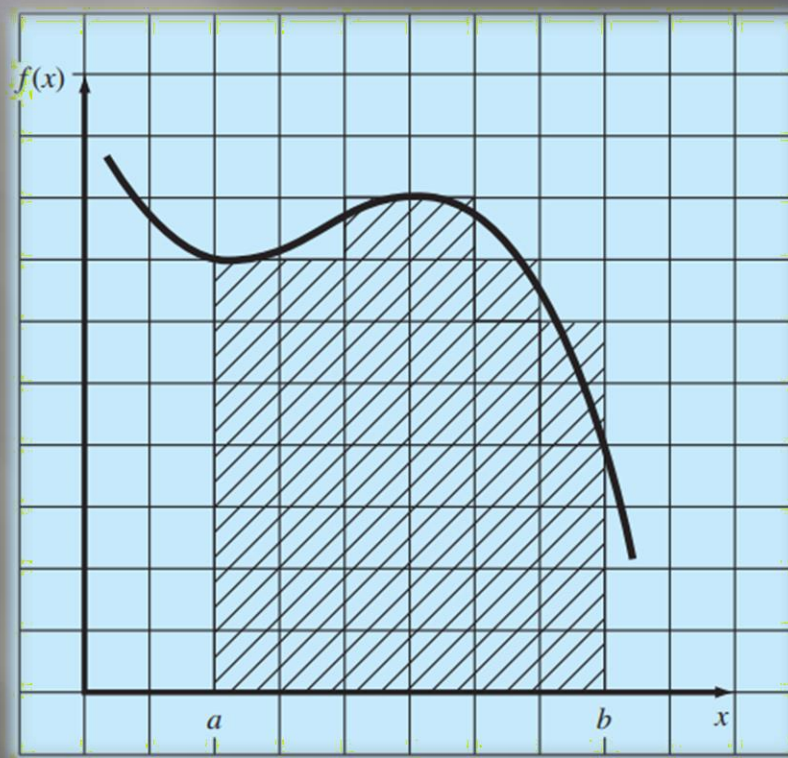
Representing the area if f is continuous and nonnegative on the closed interval $[a, b]$



Definite Integral (2/13)

Examples of Numerical Solutions (1/5):

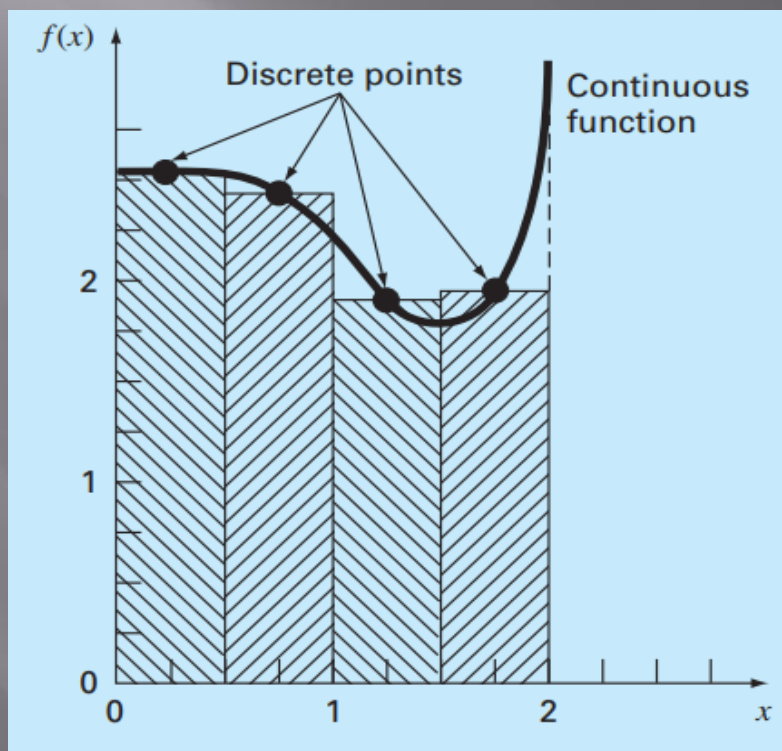
The use of a grid to approximate an integral.



Definite Integral (2/13)

Examples of Numerical Solutions (2/5):

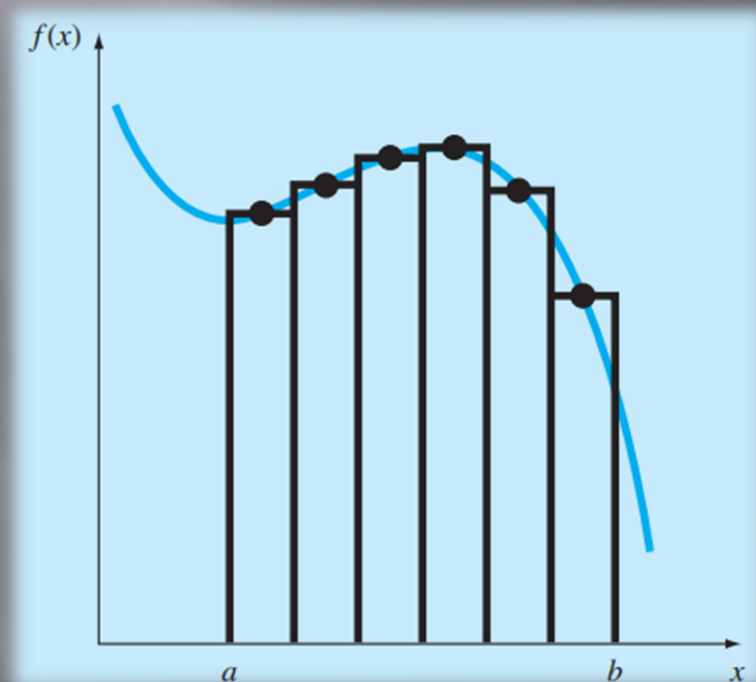
The use of rectangles, or strips, to approximate the integral.



Definite Integral (2/13)

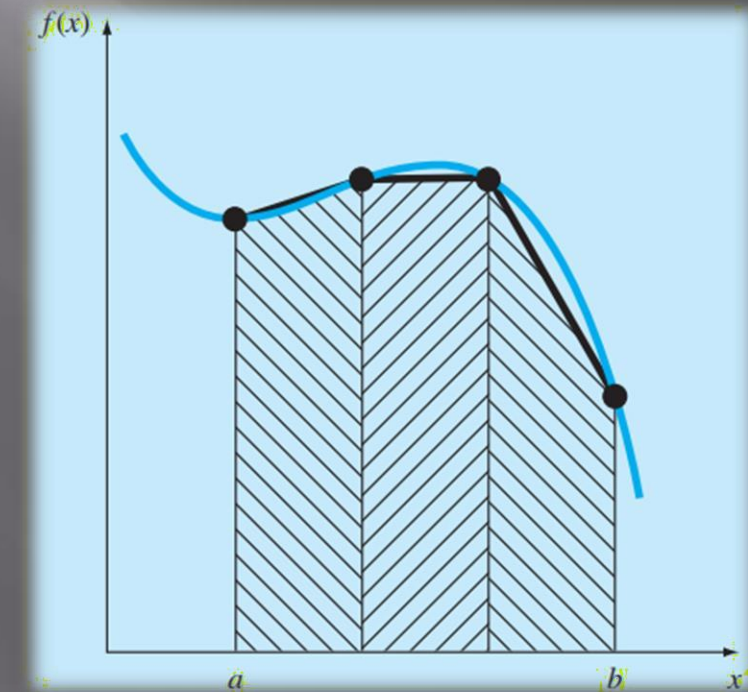
Examples of Numerical Solutions (3/5):

The use of rectangles, or strips, to approximate the integral.



Definite Integral (2/13)

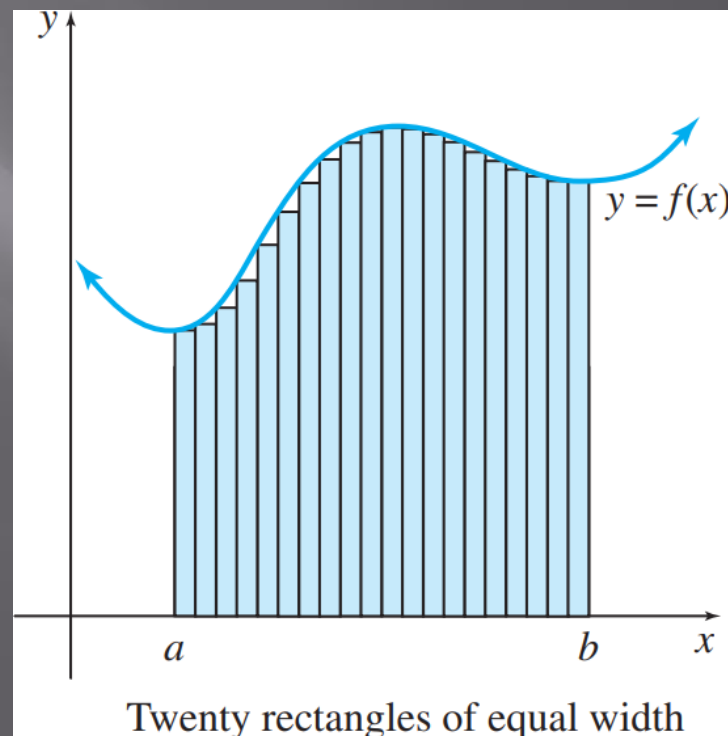
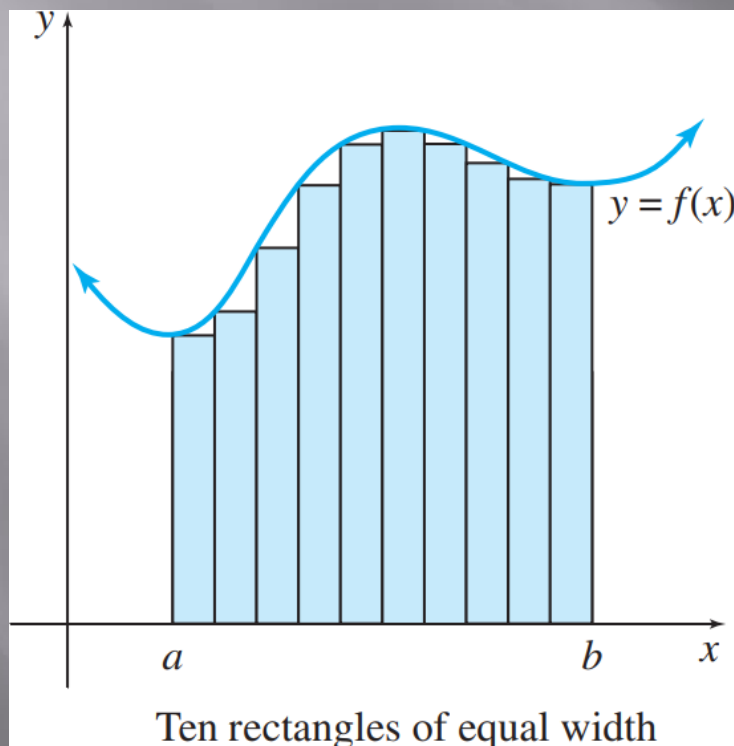
- ▣ Examples of Numerical Solutions (4/5):
- ▣ The approximation of an integral by the area under three straight-line segments or trapezoidal.



Definite Integral (2/13)

Examples of Numerical Solutions (5/5):

Different numbers of rectangles.



Definite Integral (3/13)

▣ Riemann Sum:

- ▣ Let f be defined on the closed interval $[a, b]$, and let Δ be a

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

where Δx_i is the width of the i th subinterval

$$[x_{i-1}, x_i].$$

If c_i is *any* point in the i th subinterval, then the sum

$$\sum_{i=1}^n f(c_i) \Delta x_i, \quad x_{i-1} \leq c_i \leq x_i$$



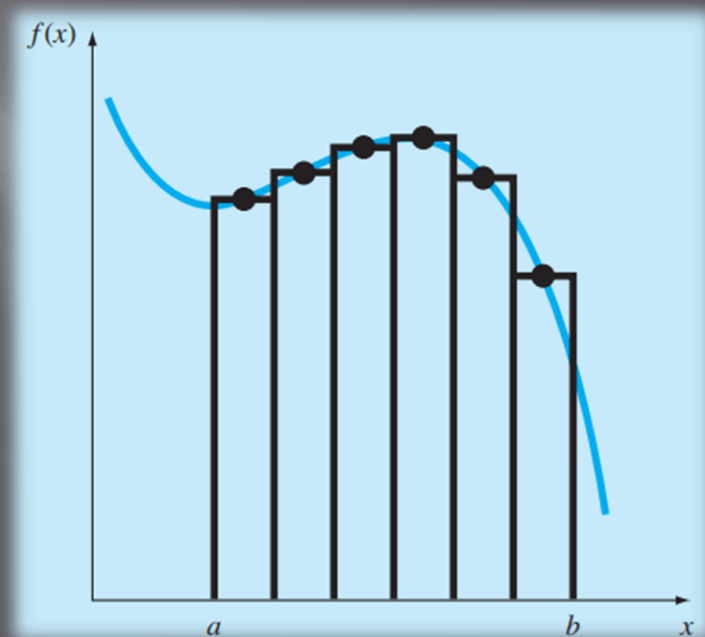
Georg Riemann
“Germany”

Definite Integral (4/13)

- ▣ Definition of Definite Integral:
- ▣ Let f be defined on the closed interval $[a, b]$, and let Δ be a partition of $[a, b]$

$$\lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i = \int_a^b f(x) dx.$$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x$$



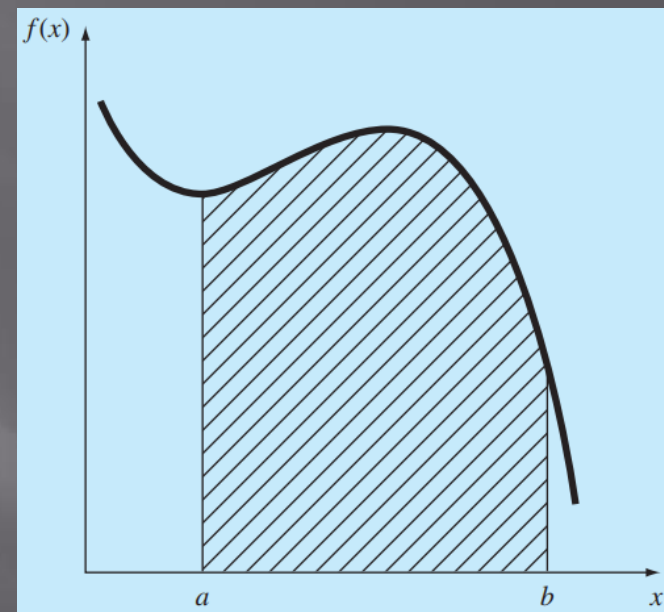
Definite Integral (5/13)

How to find the Definite Integral?

$$I = \int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$$

where F is an antiderivative of f

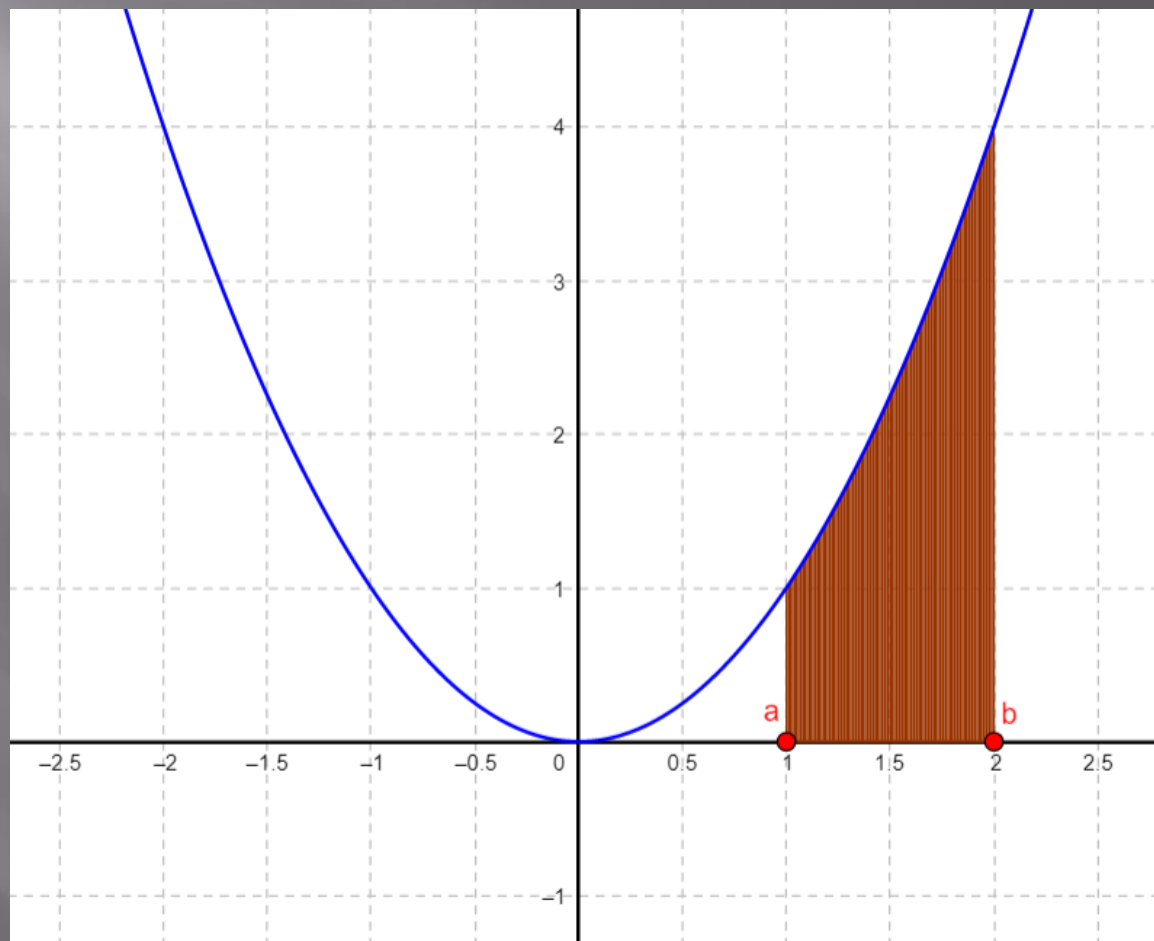
Representing the area if f is continuous and nonnegative on the closed interval $[a, b]$



Definite Integral (6/13)

Example 1 (1/2):

$$\int_1^2 x^2 dx$$



Definite Integral (6/13)

Example 1 (2/2):

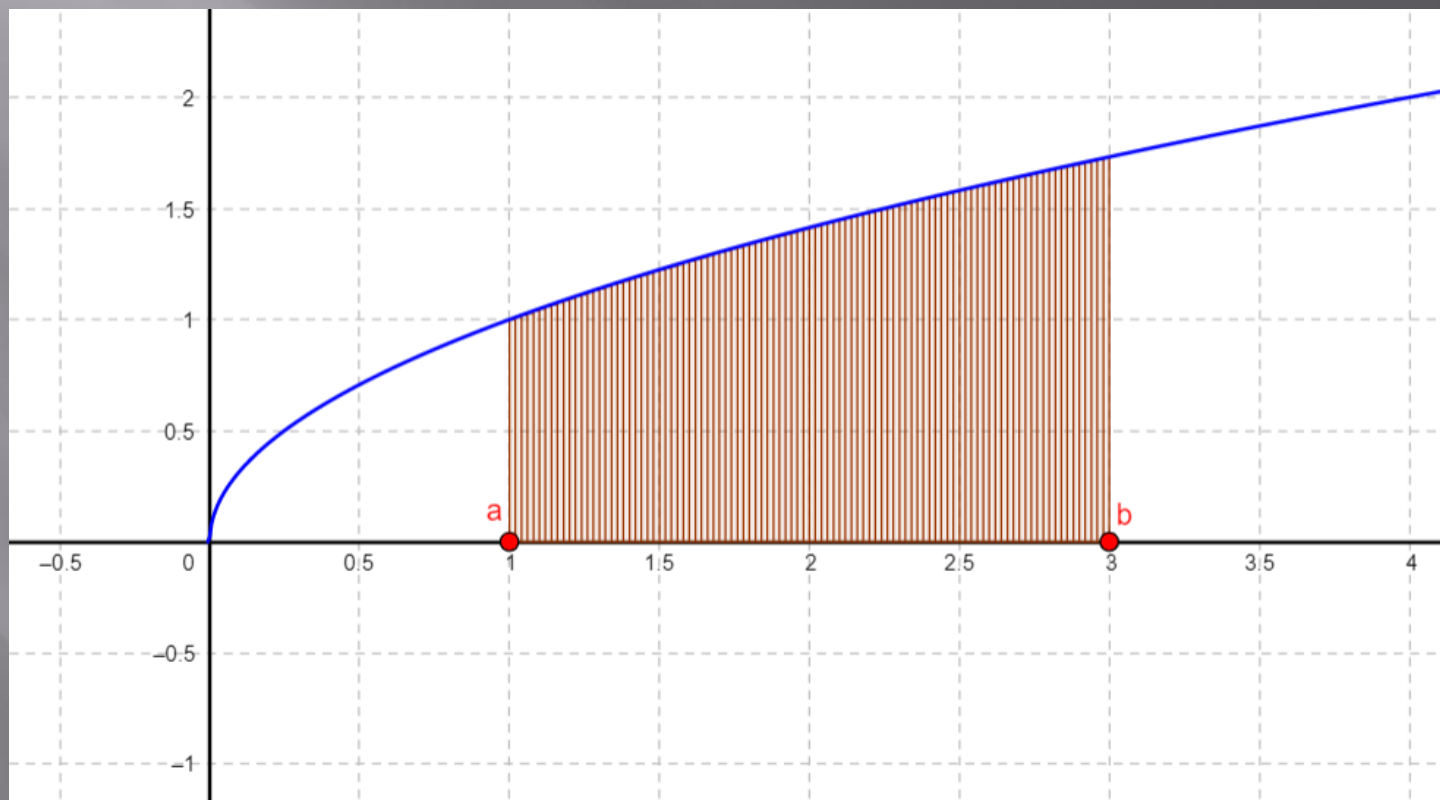
$$\int_1^2 x^2 dx$$

$$\int_1^2 x^2 dx = \left[\frac{x^3}{3} \right]_1^2 = \left[\frac{2^3}{3} \right] - \left[\frac{1^3}{3} \right] = \left[\left(\frac{8}{3} \right) - \left(\frac{1}{3} \right) \right] = \frac{7}{3} \approx 2.333$$

Definite Integral (7/13)

Example 2 (1/2):

$$\int_1^3 \sqrt{x} \, dx$$



Definite Integral (7/13)

Example 2 (2/2):

$$\int_1^3 \sqrt{x} \, dx$$

$$\int_1^3 x^{1/2} \, dx = \frac{x^{3/2}}{3/2} \Big|_1^3 = \left(\frac{(3)^{3/2}}{3/2} \right) - \left(\frac{(1)^{3/2}}{3/2} \right) = 2.797435$$

Definite Integral (8/13)

Example 3 (1/3):

1

$$\int_2^1 2x \, dx$$

-2

Definite Integral (8/13)

Example 3 (2/3):

$$\int_{-2}^1 2x \, dx = x^2 \Big|_{-2}^1 = ((1)^2) - ((-2)^2) = 1 - 4 = -3$$

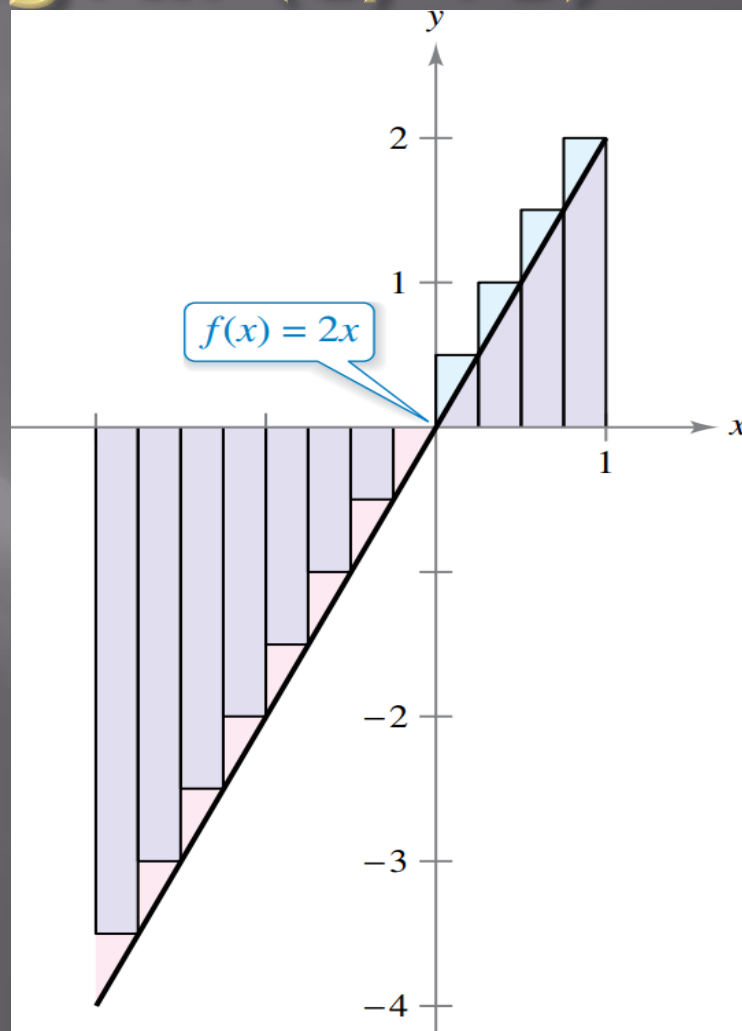
Definite Integral (8/13)

Example 3 (3/3):

1

$$\int_{-2}^1 2x \, dx = -3$$

-2



Properties of Definite Integrals (1/5)

$$\int_a^b f(x)dx = - \int_b^a f(x)dx$$

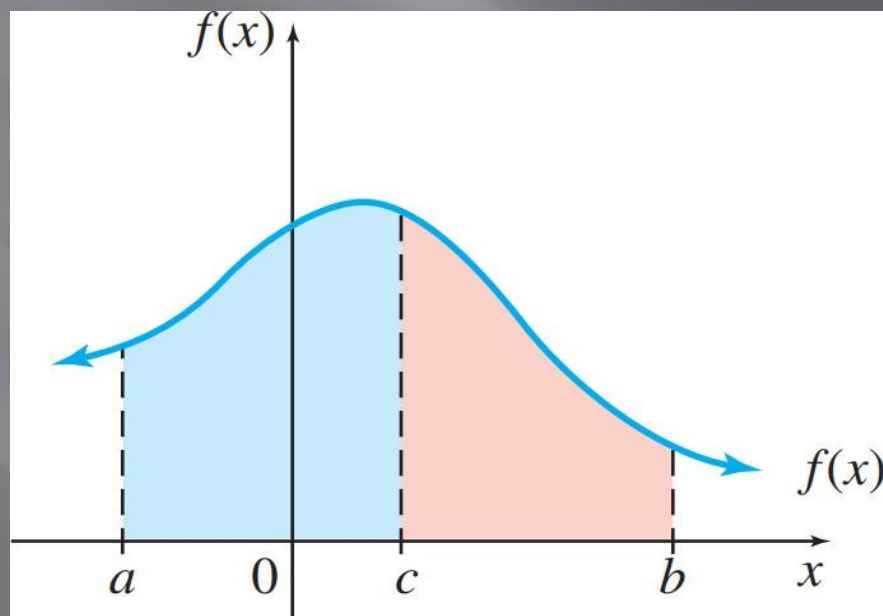
$$\int_a^a f(x)dx = 0$$

TE INTEGRAL
(9/13)

DEFINITE INTEGRAL

Properties of Definite Integrals (2/5)

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$



DEFINITE INTEGRAL (10/13)

Example 1 (1/2)

Given that $\int_0^6 f(x)dx = 8$ and $\int_5^6 f(x)dx = 7$, find $\int_0^5 f(x)dx$.

DEFINITE INTEGRAL (10/13)

Example 1 (2/2)

Given that $\int_0^6 f(x)dx = 8$ and $\int_5^6 f(x)dx = 7$, find $\int_0^5 f(x)dx$.

$$\int_0^6 f(x)dx = \int_0^5 f(x)dx + \int_5^6 f(x)dx.$$

$$8 = \int_0^5 f(x)dx + 7,$$

$$\text{and so, } \int_0^5 f(x)dx = 1$$

Thank You

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